

Product Relatedness in Tasks

A Novel Measure and Multi-Product Firms

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Motivation

Question. Do multi-product firms expand their product scope toward products whose task requirements overlap with the labor capabilities already embodied in their incumbent workforce?

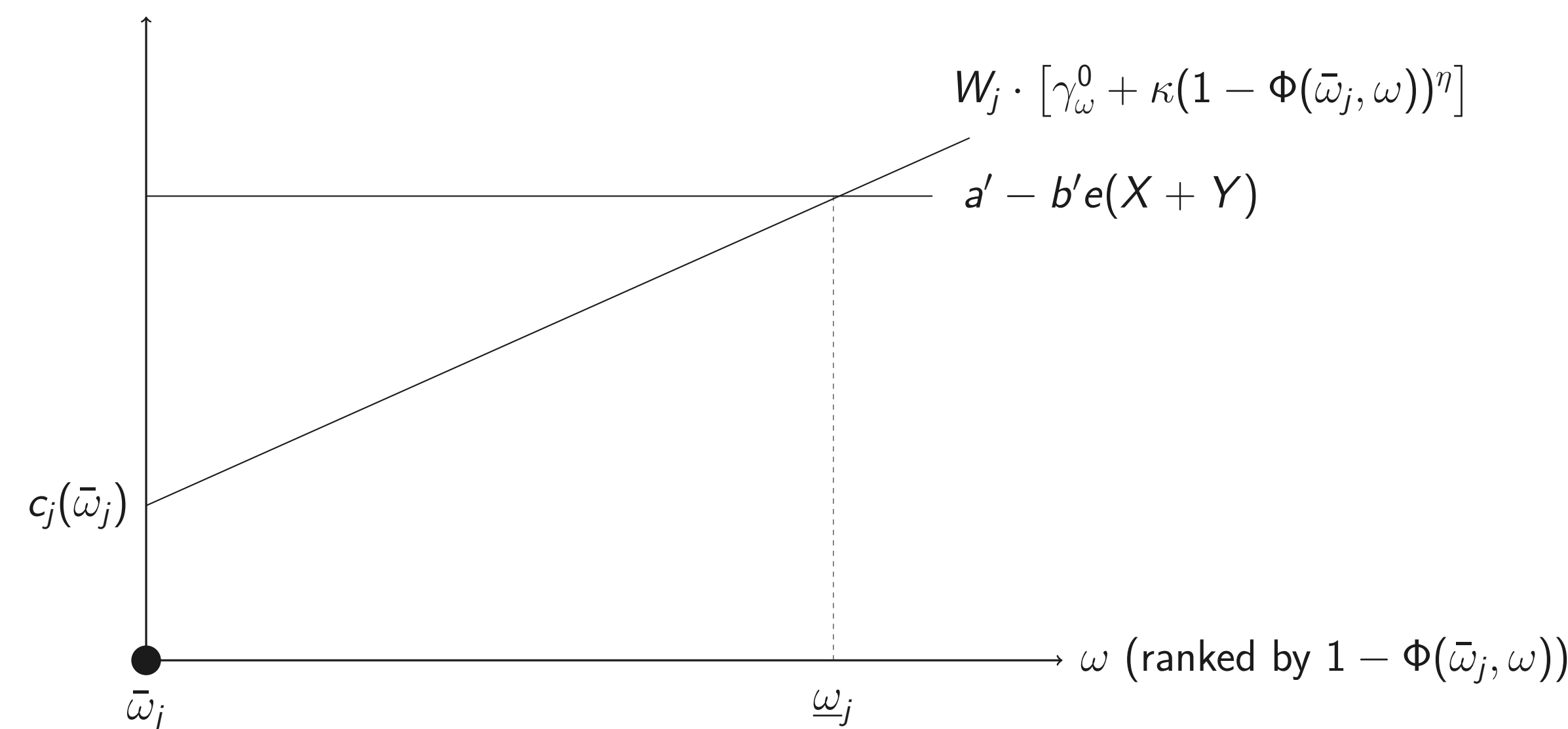
Idea. Products differ in the **occupations / tasks** they require, and firms expand by exploiting incumbent and available resources:

- 1) adding probability increases in task similarity, and
- 2) dropping probability decreases in task similarity.

Framework (Eckel & Neary, 2010)

Flexible manufacturing: firms add varieties with minimal adaptation.

The marginal cost of a candidate ω rises with its task distance to the core $\bar{\omega}_j$



Data

Source Danish employer–employee registry data, 2010–2019.

Sample $\approx 4,000$ manufacturing firms ($\approx 1/3$ being MPF);
sales at HS 4-digit (ex. "0401: Milk and cream (unsweetened)");
 $\approx 370,000$ workers in DISCO 3-digit occupations
(ex. "411: Regular office work").

Events Adding = new product in t , absent in $t - 1$; dropping is the reverse.

Stylized facts

Targeted hiring

Employment grows +22% vs. products +12%;
occupations only +5%.

Scale and scope

Within firms: more products $\rightarrow$ more workers ($\approx 5.5\%$),
not more occupations. Both increase with revenue.

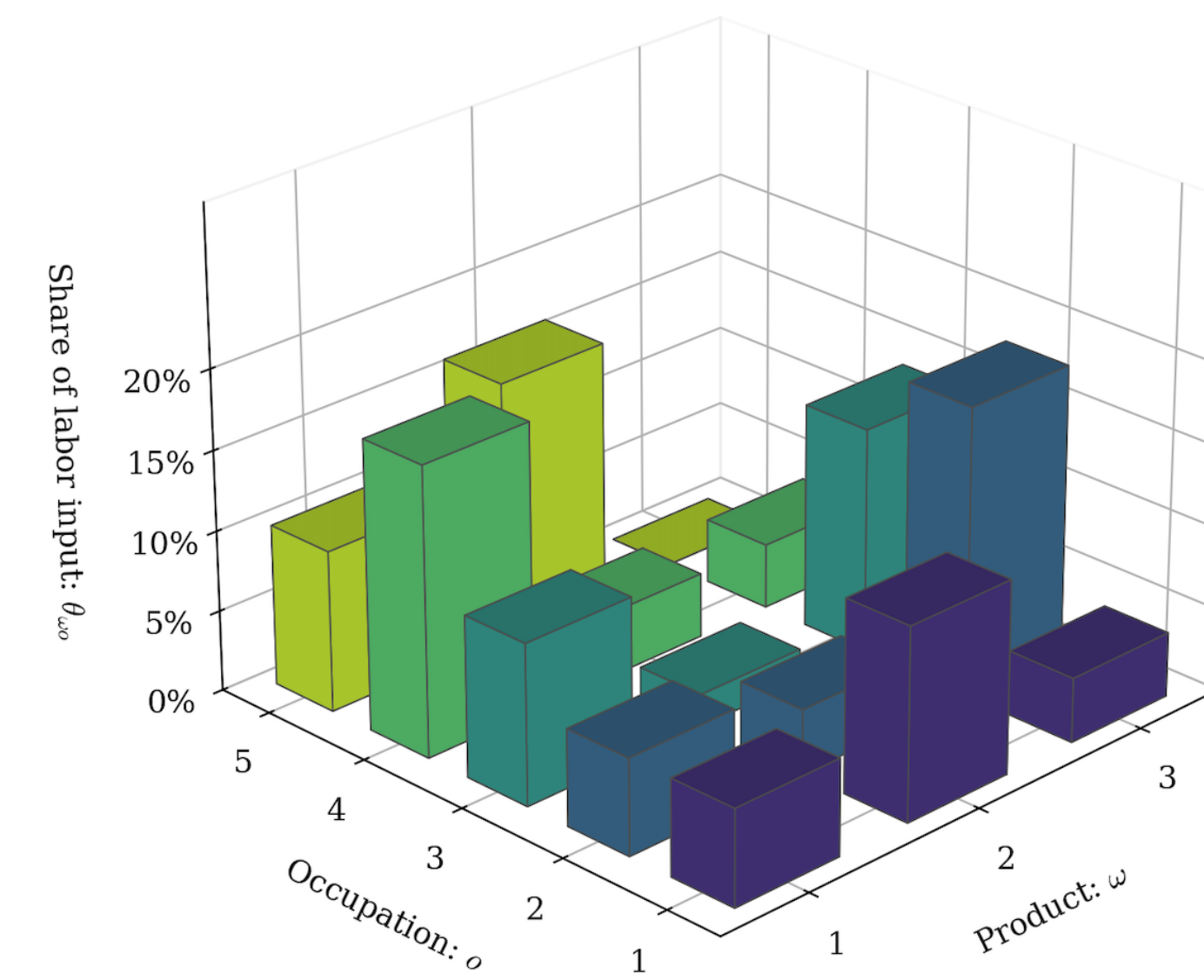
Churn rates

Higher product adding rate increases both occupational
adding rate *and* occupational dropping rate, but it only
increases worker adding rate.

Task vectors from single-product firms

SPFs. Proxy for required tasks to produce ω . Product-level vector:

$$\theta_{\omega o} = \frac{h_{\omega o}}{h_{\omega}}, \quad h_{\omega o} = \sum_{j \in \text{SPF}_{\omega}} h_{jo}$$



Weighted cosine similarity

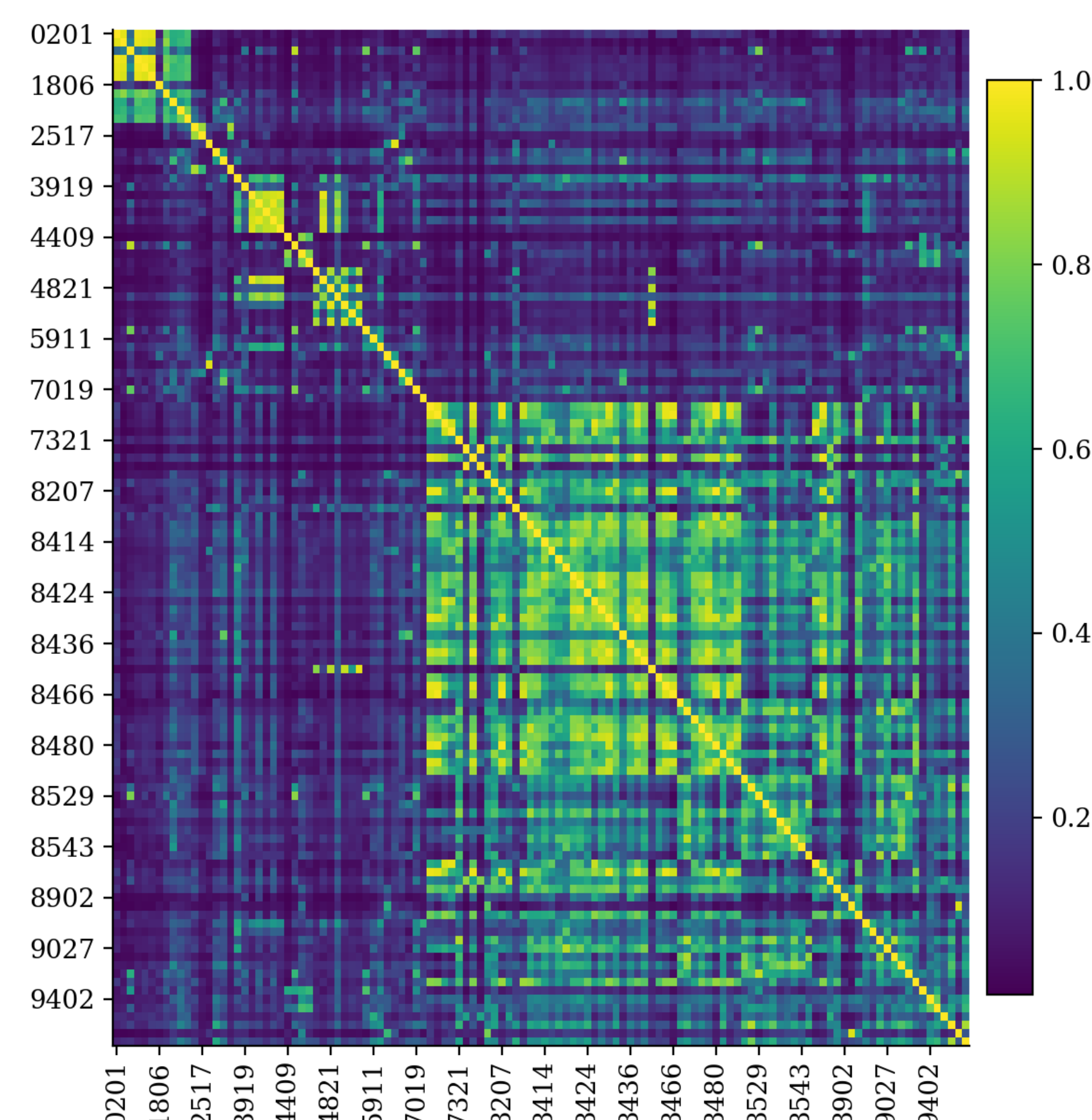
Extension. Weight *all* bilateral occupation pairs by skill transferability Ψ , not just like-for-like:

$$\Phi(\omega, \omega') = \frac{\sum_{o, o^*} \Psi_{o, o^*} \theta_{\omega o} \theta_{\omega' o^*}}{\sqrt{\sum_{o, o^*} \Psi_{o, o^*} \theta_{\omega o} \theta_{\omega o^*}} \sqrt{\sum_{o, o^*} \Psi_{o, o^*} \theta_{\omega' o} \theta_{\omega' o^*}}}$$

Weights. Excess worker flows (Neffke & Henning, 2013) via a ZINB model:

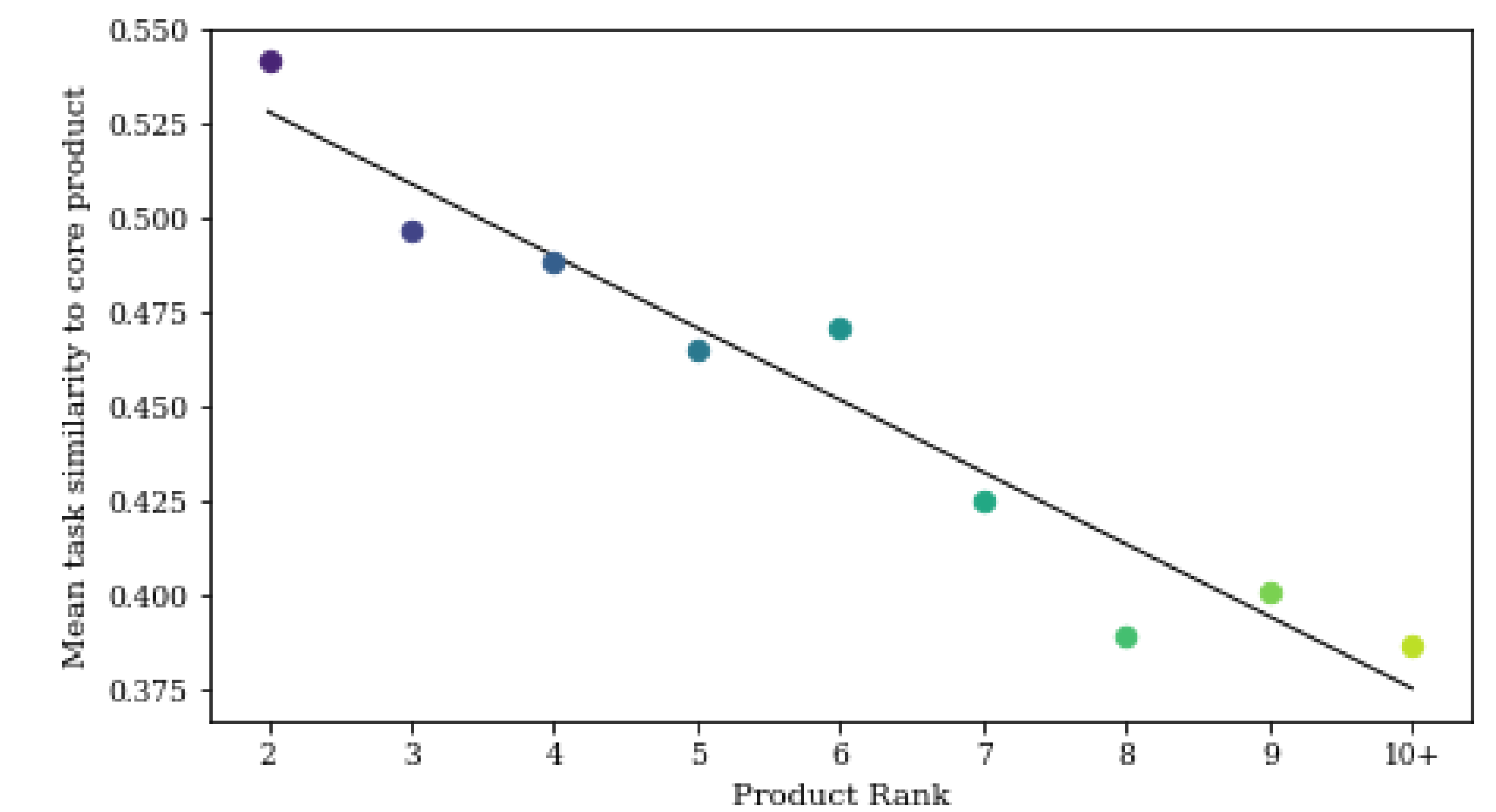
$$\Psi_{o, o^*} = F_{o, o^*} / \hat{F}_{o, o^*}$$

Clusters. Food, plastics & rubber, and a large machinery core.

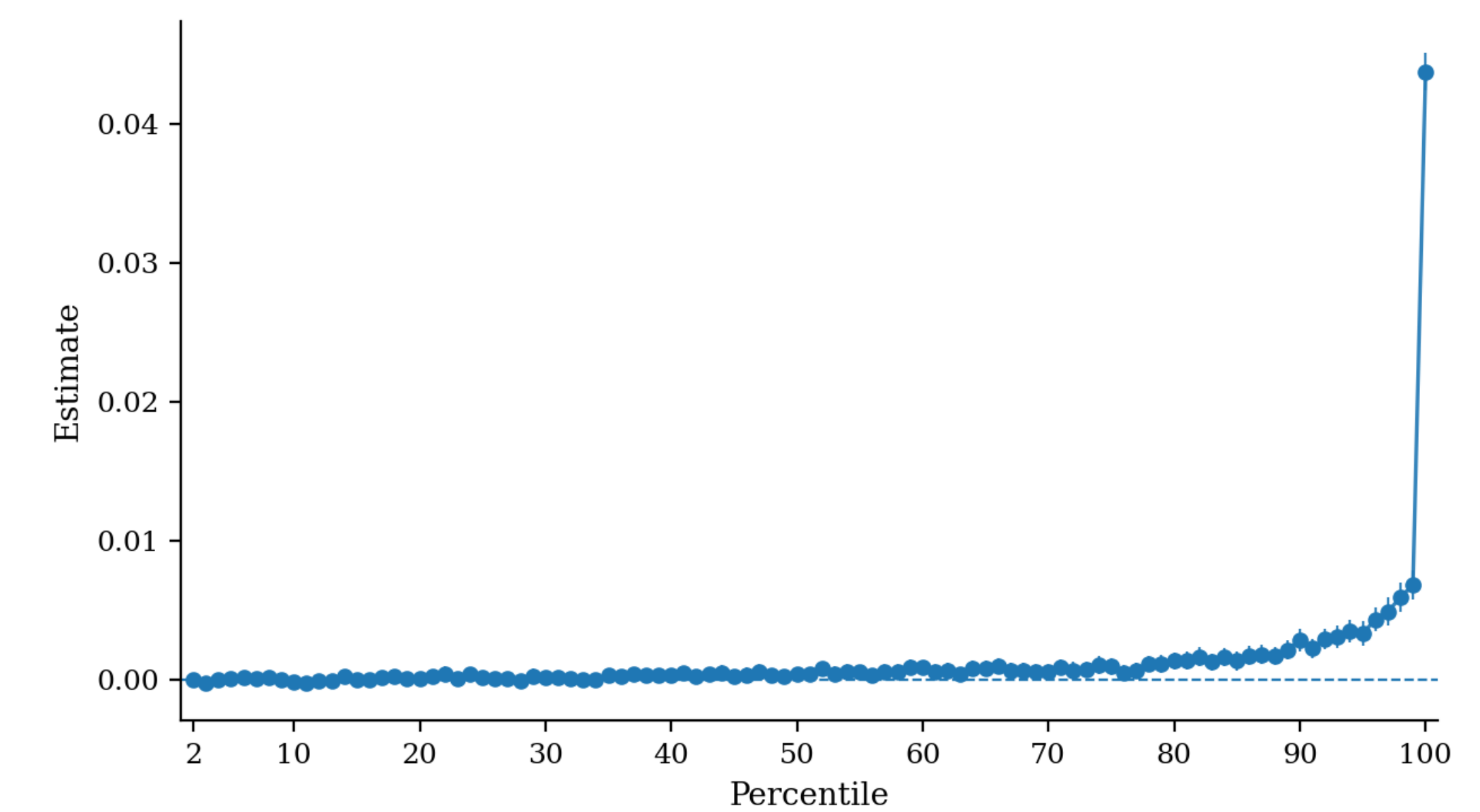


Multi-product firms: adding & dropping

Adding near core competencies. Similarity to the core falls with a product's sales rank:



Adding probability. Increases with +1.33pp*** per unit of Φ ; concentrated in top percentiles:



Dropping probability. Decreases with 6.30pp*** per rank; rises for low-rank, distant products.

IV strategy (in progress)

Shift-share. Shift = non-Danish import demand $\Delta_{\omega, t}^{\text{NDIM}}$; share = task overlap (adding) or inverse sales share (dropping):

$$Z_{jt}^{\text{Add}} = \sum_{\omega \notin \Omega_{j, t-1}} \Phi(\bar{\omega}_{jt}, \omega) \cdot \Delta_{\omega, t}^{\text{NDIM}},$$
$$Z_{jt}^{\text{Drop}} = \sum_{\omega \in \Omega_{j, t-1}} (1 - \theta_{j\omega, t-1}) \cdot (-\Delta_{\omega, t}^{\text{NDIM}}).$$

Outcomes: revenue / VA per worker, hiring and firing, reorganization.

Takeaways

Observed-data measure. SPF vectors recover product-level task content.

Meaningful structure. Weighted cosine reveals sparse, cross-cutting clusters.

Core competencies related growth. MPFs expand along task similar paths.